

Video Communication and Virtual Collaboration Platform

1. Project Overview & Objectives

Secure Communication

WebRTC-based rooms with passcodes, waiting rooms, and transport-level controls to protect session integrity.

MFA Security

Demonstration MFA flow at login (mock verification) to validate enterprise authentication patterns.

Collaborative Workspaces

Persistent whiteboards, shared markdown notes, chat threads, polls, and Q&A inside each meeting.

AI-assisted Analytics

Live transcription, translation, searchable summaries, and automated action-item extraction.

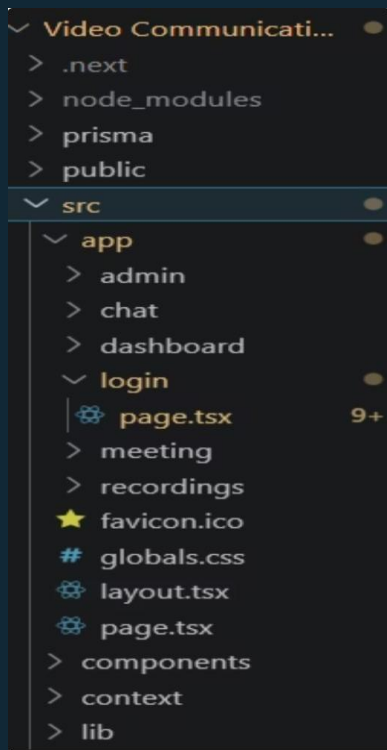
Recording Vault

Browser MediaRecorder captures sessions with interactive transcripts that jump to timestamps.

Enterprise Control

Custom SVG charts for usage, license allocation, and audit-log analysis.

2. Folder Structure & Purpose



Root — project metadata and build config (package.json, tsconfig, next.config).

prisma/ — schema.prisma defines relational models and local SQLite engine (prisma/dev.db).

public/ — static assets: logos, SVGs, icons used across the app.

src/app/ — Next.js App Router pages and API routes; key routes: admin, chat, dashboard, login, meeting/[id], recordings.

src/components/ — reusable widgets: Sidebar, Whiteboard, Notes, ChatPanel, PollsPanel, QAPanel, AIAssistant, AdminCharts.

src/context/ — AppContext manages sessions, user roles, and global state.

src/lib/ — prisma.ts exports the initialized Prisma client for server-side data access.

3. Step-by-Step Local Runbook

Follow this concise runbook to reproduce the app locally. Each command runs from the project root.

1 Step 1 — Open Terminal

```
cd "C:\\Users\\ravik\\OneDrive\\Desktop\\VC2\\Video Communication Platform"
```

2 Step 2 — Install Dependencies

`npm install` — installs next, react, prisma, tailwind, and other dependencies declared in `package.json`.

3 Step 3 — Initialize Database

```
npx prisma generate
```

```
npx prisma db push
```

 — generates client and pushes schema to local SQLite (`prisma/dev.db`).

4 Step 4 — Start Dev Server

```
npm run dev
```

 — boots the Next.js development server and hot-reloads routes.

5 Step 5 — Open App

Open `http://localhost:3000` in a modern browser (Chrome/Edge/Firefox/Safari with WebRTC support).

Tip: For PostgreSQL, update `prisma/schema.prisma` `datasource` and add `DATABASE_URL` to `.env` before running `prisma generate` & `db push`.

4 Step by Step Running Steps

Initialize Database

```
PS C:\Users\ravik\OneDrive\Desktop\VC2\Video Communication Platform> npx prisma generate
>> npx prisma db push
```

```
Update available 6.19.3 -> 7.8.0
```

```
This is a major update - please follow the guide at
https://pris.ly/d/major-version-upgrade
```

```
Run the following to update
```

```
npm i --save-dev prisma@latest
```

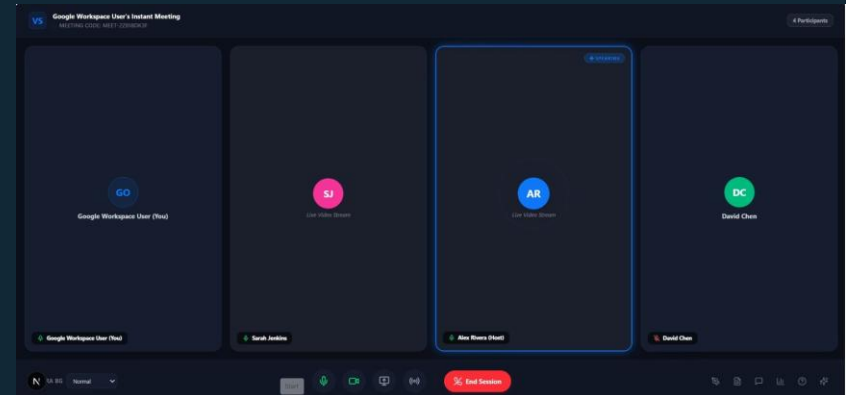
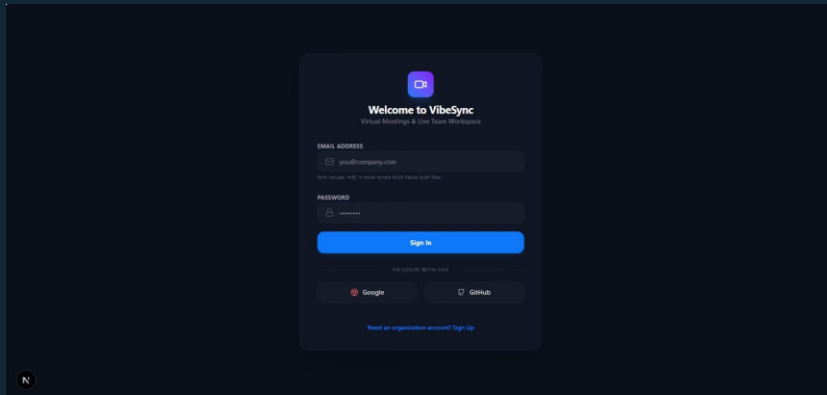
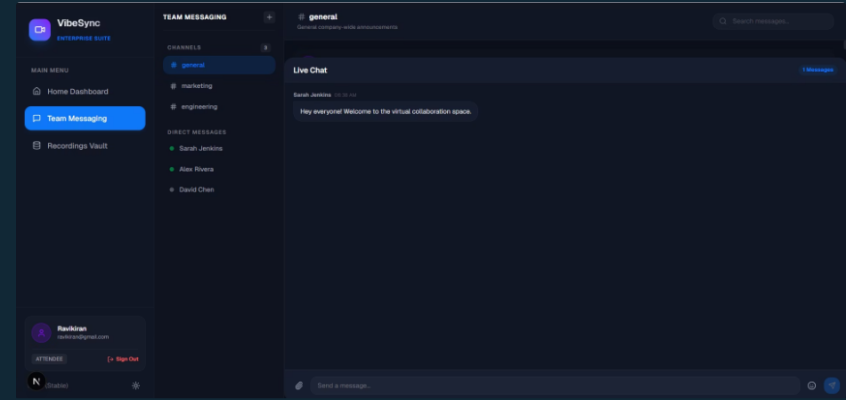
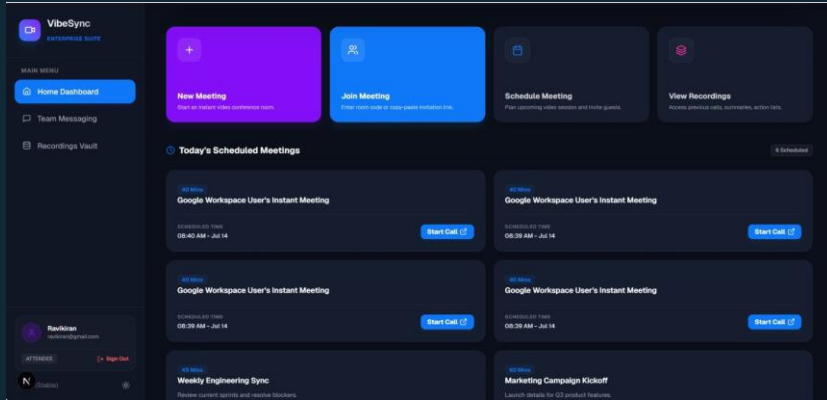
```
npm i @prisma/client@latest
```

Start Dev Server

```
mmunication Platform> npx prisma generate
```

```
mmunication Platform> npm run dev
```

5. Execution Process & Output Screenshots (MANDATORY)



6. Required Software, Dependencies & Additional Notes



Runtime

Node.js v18+ (LTS recommended).



Package Manager

NPM (bundled with Node). Use `npm ci` for consistent installs in CI.



Browser

Chrome/Edge/Firefox/Safari with WebRTC & getUserMedia support.

Key Packages (from package.json)

- next v16.2.10 — App Router and server rendering
- react / react-dom v19.2.4 — UI framework
- prisma / @prisma/client v6.2.1 — schema and query client
- tailwindcss v4 — utility-first styling



Testing MFA: use an email containing "mfa" to trigger mock passcode flow. Use code 123456 to pass verification.